LEEDS CITY COUNCIL

Highways and Transportation

Highways Services
Civic Hall
Portland Way
Leeds LS1 1UR

Junction:	Hyde Park Road Toucan near Brudenell Road		
Company:	Dynniq	Controller Type:	PTC1
Mains Supply:	230V / 50Hz	New / ~~Modification~~	
Operating Voltage:	ELV		
Engineer:	Adam Gudgeon	Telephone No:	07566 298199

OTU SCN	03450	Cont SCN	03451	MOVA SCN	03452

IP Address Information:

Controller IP:	xxx.xxx.xxx.xxx	Gateway:	xxx.xxx.xxx.xxx
Mask:	xxx.xxx.xxx.xxx	OTU IP:	xxx.xxx.xxx.xxx

NOTE TO UTC PERSONNEL — This specification should be used for general purposes only. The Signal Company documentation should be used for controller operation and timings unless amended on this sheet as RAM data.

RAM UPDATE (If no data then write NONE)	Reasons for change	Initials	Date
Write date current EPROM fitted:-			

ORN: 397L	Scheme Number:	Issue: 1	Date: 04/12/2024

397L Hyde Park Road Toucan near Brudenell Road

Configuration Notes

A telent Optima controller shall provide the method of control in this specification.

UTC control will be via the Optimas UG405 inerface.

The controller shall be capable of switching between normal and pre time max operation. By default it shall operate pre time max but this facility may be suspended by a software switch.

A clearly labelled, manually selected VA switch shall be provided within the manual panel which shall override lower priority modes when active. It shall also be possible to force the controller into selected VA via the controller web pages.

Audible Signal
An Audible signal shall be persent during the green man period with the facility to be switched by timetable.
Initially set to turn off the audible signal everyday 22:00-07:00.

VA Pre-timed max with additional timer
There will be TWO max green timers. The first will be pre-timed. The 2nd VA max timer (DA), set to 5 seconds will start upon receipt of a ped demand. The pre timed max will commence after the start of vehicle green and will run for 20 seconds. If durning this time a pedestian demand is registered then gap change will be permitted, should no vehicle extensions be present. After the per timed max has expired and there has been no pedestrain demand the second the second VA max time will govern operation. If a demand for the pesetrain is recieved during the per-timed max period tehre is still some residue of the second VA max then the second max will govern operation and continue to extend the traffic phase. If the second VA max has expired at the end of the per-time max then the pedestrian demand will be activated.

Kerbisde Detector Operation
The Kerbside detectors will operate in 3 ways:

1. To eliminate the requirement for pedestrians to place a demand usning the push button the kerbside detectors shall place an unlatched demand for pedestrian phase C, initially configured with a call delay of 2 seconds and a cancel period of 2 seconds (alterable by handset). This operation shall be configured with software switch that can be switched by timetable and initially set to **incative.**

2. The general operation of the kerbside and pushbutton demands will be as set out in TOPAS 2500. Pushbutton demands will be held and cancelled depending on the active state of the kerbside and any push button demand placed when no kerbside presence is detected shall place a latched demand.

3. The raw output from the kerbside shall set MOVA det 41. This shall be used to extend the green man period through the use of period links within MOVA.

NOTE: TBC - On DFM fail the kerbside detectors will not insert a demand for phase C and shall not set MOVA det 41. At this time a pushbutton demand will be required.

If unlatched demand is placed for phase C and active kerbsides for that phase will hold the demand; the demand will only be dropped if all kerbsides associated with the phase are inactive.

ORN:	397L	Scheme Number:	0	Issue:	1	Date:	04/12/2024

Configuration Notes

MOVA Conditioning

MOVA UTC Interface stream 1:
Control: F1 F2 MTO1
Reply: G1 G2 CRB1 MO1
Wait Lamp Mimic: Whilst pedestrian phase wait lamp is lit, input to MOVA:
Wait Confirm Ped c = MOVA Det 13
CRB Toggle: CRB configured to toggle every 120 seconds if MOVA is not in control

The following control bits are processed by UTC and sent based on different traffic conditions on the corridor. The purpose of these is to hold the crossing at green for vehicles to flush traffic through when required.

If control bit SF1 is received, pulse MOVA Det 62 Stream 1 for 2 seconds. (44578)
If control bit SF1 is received, pulse MOVA Det 63 Stream 1 for 2 seconds. (44579)

Loop positions
Detectors shall be wired into packs as shown bellow to prevent chatteing.
To assist in Validation all detector inputs to be clearly labelled on detetor packs in the controller.

Approach	loop	Dist From Stop Line (M)	Longitudual demension (M)	Shape	Clearance distance		Det Pack No.
					N/S	O/S	
Hyde Park Road Northbound	AX1	40	1.7	Dimond	0.4	0.8	1
Hyde Park Road Southbound	BIN2	80	1.7	Dimond	0.4	0.8	1
	BX3	40	1.7	Dimond	0.4	0.8	1

Configuration Notes

Corridor Control Strategy

The following control bits are processed by UTC and sent based on different traffic conditions on the corridor. The purpose of these is to hold vehicle green to flush traffic through when required.

If control bit 17 (SF1) is received, pulse MOVA Det 29 for 2 seconds. (SCN 44574)
If control bit 18 (SF2) is received, pulse MOVA Det 30 for 2 seconds. (SCN 44575)
If control bit 19 (SF3) is received, pulse MOVA Det 31 for 2 seconds. (SCN 44576)
If control bit 20 (SF4) is received, pulse MOVA Det 32 for 2 seconds. (SCN 44577)

Dynamic Dataset Changes

Based on the flow/occupancy/queueing conditions of the corridor it is desirable to change MOVA Datasets. The following logic and MOVA detectors facilitate the changes.

If control bit 12 (DS1) is being received from UTC, output to MOVA Det.25 (SCN 44570)
If control bit 13 (DS2) is being received from UTC, output to MOVA Det.26 (SCN 44571)
If control bit 14 (DS3) is being received from UTC, output to MOVA Det.27 (SCN 44572)
If control bit 15 (DS4) is being received from UTC, output to MOVA Det.28 (SCN 44573)

Flow / Occupancy Loops

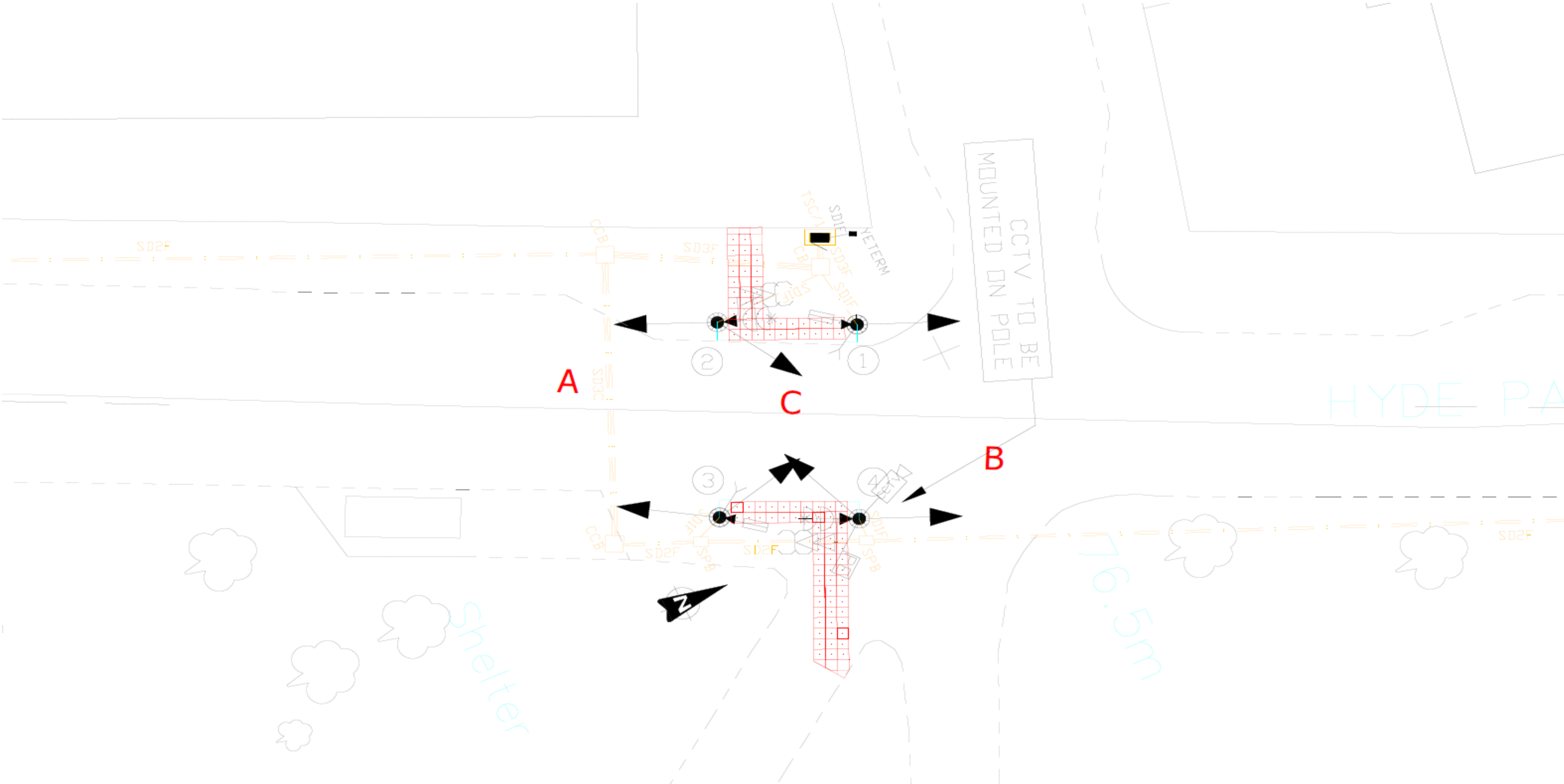
To monitor the variouse profiles of vehicles passing through the junction the following conditioning is required on each of the detection loops listed bellow:

AX1 - (SCN - 44578)
• The raw state of AX1 will output to mova det 1, stream 1
• The raw state of AX1 will output to UTC via OC1
• Every 4th vehicle passing over AX1 will toggle the state of FL1

BIN2 - (SCN - 44579)
• The raw state of BIN2 will output to mova det 2, stream 1
• The raw state of BIN2 will output to UTC via OC2
• Every 4th vehicle passing over BIN2 will toggle the state of FL2

ORN: 0	Scheme Number: 0	Issue: 1	Date: 00/01/1900

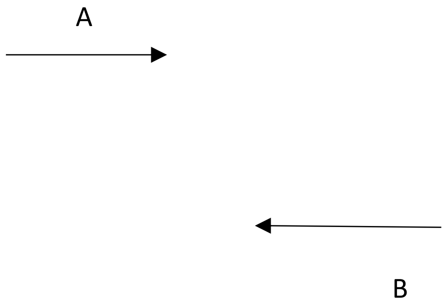
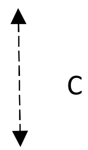
Site Layout



NOT TO SCALE

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

BASIC STAGE DATA FOR STREAM NO 1

Stage: 1	Stage: 2	Stage: 3	Stage:
		All Red Stream 1	
Stage:	Stage:	Stage:	Stage:

STREAM NO.	1	2	3	4	5	6	7	8
ALL RED STAGE	0							
START UP STAGE	0							
REVERSION STAGE	0							

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

	USE OF PHASES	PHASE	APPEAR	TERMINATION					MAXIMUMS			
									(Day)	(AM)	(PM)	(Night)
	LOCATION	TYPE NOTE 3	TYPE NOTE 1	TYPE NOTE 2	ASSOC PHASE	PED BLKT	PERIOD V	MIN GREEN	SET A	SET B	SET C	SET D
A	Hyde Park Road Northbound	T	0	0				7				
B	Hyde Park Road Southbound	T	0	0				7				
C	Pedestrian	PU	0	0		-	3	6				
D	Pre-time Max Dummy	D	0	0				1				

Note 1:- Condition of Appearance

0 - Phase always appears when stage runs

1 - Phase only appears if demand exists at start of inter-stage

2 - Phase appears, if demanded, at any time up until the end of the stage in which the phase runs

3 - Phase appears, if demanded at any time during the stage up until the window time expires

Note 2:- Condition of Termination

0 - Terminates at end of stage

1 - Terminates when associated phase gains right of way

2 - Terminates when associated phase loses right of way

3 - Terminates at end of minimum green

4 - Terminates at end of maximum green

5 - Terminates subject to special conditioning

Note 3: - Phase Types

T - Traffic	F - Filter Arrow
PD- Pedestrian	I - Indicative Arrow
PU- Puffin	D - Dummy
TN - Toucan Nearside	L - L.R.T
TF - Toucan Farside	S - Switched Sign
	C - Cycle

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

		Phase Maxima							
		Max A	Max B	Max C	Max D	Max E	Max F	Max G	Max H
A	Hyde Park Road Northbound	20	15	20	15	15	15	15	15
B	Hyde Park Road Southbound	20	15	20	15	15	15	15	15
C	Pedestrian								
D	Pre-time Max Dummy								

ORN:- 397L	Scheme Number:- 0	Issue:- 1.00	Date:- 04/12/2024

PUFFIN CROSSING TIMINGS

Phase	Crossing Width (m)	Comfort Factor	Period III	Period IV	Period V	Period VI	Period VII	Period VIII
C	7	3	3	6	3	9	0	0

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

USE OF STAGES - STAGES/PHASE RELATIONSHIP

STAGE	A	B	C	D																			
1	✓	✓																					
2			✓																				
3																							
4																							
5																							
6																							
7																							
8																							
9																							
10																							
11																							
12																							
13																							
14																							
15																							
16																							
17																							
18																							
19																							

STREAM	WINDOW TIME
1	
1	
1	

MAX REVERTIVE DEMAND PHASE	✓	✓																					

✓ *Tick appropriate boxes*

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

PHASES			CONFLICTING / OPPOSING PHASES																				
	A	B	C	D																			
A			C																				
B			C																				
C	C	C																					
D																							

C = CONFLICTING & OPPOSING

O = OPPOSING ONLY

. = NO CONFLICT

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

PHASES			PHASE INTERGREEN TIMES																				
	A	B	C	D																			
A			6																				
B			6																				
C	5	5																					
D																							

STARTING INTERGREEN PERIOD	15

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

PHASES	PHASE INTERGREEN HANDSET LIMIT VALUES																						
	A	B	C	D																			
A			5																				
B			5																				
C	5	5																					
D																							

MIN GREEN VALUE	
LOW	HIGH
7	255
7	255
4	255

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

ADDITIONAL PHASE DELAYS														
No	Delay Phase	Change From	To Stage	By (X) Secs	No	Delay Phase	Change From	To Stage	By (X) Secs	No	Delay Phase	Change From	To Stage	By (X) Secs
1					29					57				
2					30					58				
3					31					59				
4					32					60				
5					33					61				
6					34					62				
7					35					63				
8					36					64				
9					37					65				
10					38					66				
11					39					67				
12					40					68				
13					41					69				
14					42					70				
15					43					71				
16					44					72				
17					45					73				
18					46					74				
19					47					75				
20					48					76				
21					49					77				
22					50					78				
23					51					79				
24					52					80				
25					53					81				
26					54					82				
27					55					83				
28					56					84				

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

Red lamp monitoring

Incoming phase to be inhibited / Extinguish signals

Red Lamp failed on phase

Phase	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y			
A			✓																									
B			✓																									
C																												
D																												
E																												
F																												
G																												
H																												
I																												
J																												
K																												
L																												
M																												
N																												
O																												
P																												
Q																												
R																												
S																												
T																												
U																												
V																												
W																												
X																												
Y																												

Please note - No intergreen extension is required after first red lamp fail ✓ *Tick appropriate boxes*

ORN:- 397L	Scheme Number:-	Issue:- 1	Date:- 04/12/2024

PROHIBITED / ALTERNATIVE STAGE MOVEMENTS UP TO 16 STAGES

FROM	TO STAGE															
STAGE	1	2	3													
1																
2																
3																

TICK AS REQUIRED		
MODE	These Restrictions are for	No Restrictions
Urban Traffic Control		✓
Cableless Linking		✓
Vehicle Actuated		✓
MOVA		✓
Hurry Call		✓
Manual		✓
Priority		✓
Emergency		✓

NOTE:

✓	=	Permitted
P	=	Prohibited
X	=	Ignore
N'	=	ALTERNATIVE MOVES where 'N' is number of stage to move to

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS													
ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
1	AX1	V		A				4			1	A	MOVA DET 1
2	BIN2	V		B				5			1	A	MOVA DET 2
3	BX3	V		B				4			1	A	MOVA DET 3
4	CPB1_P1	PB	C								3	Y	MOVA DET 13
5	CPB2_P2	PB	C								3	Y	MOVA DET 13
6	CPB3_P3	PB	C								3	Y	MOVA DET 13
7	CPB4_P4	PB	C								3	Y	MOVA DET 13
8	CONX1_P1	O						0.5			2	A	
9	CONX2_P3	O						0.5			2	A	
10	CKBS1_P1	K							4 / 5	1.0	3	A	
11	CKBS2_P3	K							6 / 7	1.0	3	A	

✓ Tick appropriate boxes

Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	30	18
3	10	-
4	10	-
5	60	72

Detector Types:-

V	Vehicle	**C**	Clearance	**SU**	Speed Upstream
PB	Pushbutton	**Q**	Queue	**SD**	Speed Downstream
K	Kerbside	**U1**	Uni Directional 1		
O	On-Crossing	**U2**	Uni Directional 2		

Error States

A	Fail Active
I	Fail Inactive
Y	Fail in input state

Paired Detector:-

Specify ID of paired Uni-directional or kerbside/pushbuttons

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

MOVA DETECTORS STREAM 1								
No.	Name / Description	Action	No.	Name / Description	Action	No.	Name / Description	Action
1	AX1		30	DS1	MOVA Dataset 1	59		
2	BIN2		31	DS2	MOVA Dataset 2	60		
3	BX3		32	DS3	MOVA Dataset 3	61		
4			33	DS4	MOVA Dataset 4	62	Priority Hold A	Holds A
5			34	SF1	UTC Hold A	63	Priority Hold B	Holds B
6			35	SF2	Priority A	64		
7			36	SF3	UTC Hold B			
8			37	SF4	Priority B			
9			38					
10			39					
11			40					
12			41	Kerbside	Green Man Extension			
13	Wait Confirm C	Demand C	42					
14			43					
15			44					
16			45					
17			46					
18			47					
19			48					
20			49					
21			50					
22			51					
23			52					
24			53					
25			54					
26			55					
27			56					
28			57					
29			58					

ORN:- 435L	Scheme Number:- 0	Issue:- 1.0	Date:- 10/01/2024

EXTEND ALL RED FACILITY - (BY DETECTOR) CONSIDERED AS PART OF INTERGREENS

Stage to Stage transition for which ALL RED extension required					
UNIT No	From Stage	To Stage	Ext. Time	All Red Max	Associated Detector
1					
2					
3					
4					
5					
6					
7					

EXTEND ALL RED TIMINGS				
STREAMS	0	1	2	3
EXTENSION TIME (SECS)*				
ALL RED MAXIMUM TIME (SECS)*				

* Measured from point at which first phase in new stage would normally start its red-green transition period.

RED EXTENSION	ENTER CHOICE
MANUAL	
CLF	
UTC	
HC	
PRIORITY	
EMERGENCY	
VA	
FT	

For Choice, enter as follows:-
O = Operative
M = Extend to Maximum
N = No extensions
(Not approved for UK)

If table is left blank then all modes will be operative.

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

MODE PRIORITY

ASSIGN NUMBER 1 = TOP PRIORITY
2 = SECONDY PRIORITY etc.
X = NOT USED

	PART TIME
	EMERGENCY VEHICLE
	HURRY CALL(S)
2	URBAN TRAFFIC CONTROL
1	SELECTED MANUAL CONTROL (1
3	SELECTED FIXED TIME OR VA (2)
5	CABLELESS LINKING
4	MOVA
	BUS PRIORITY
6	EITHER VEHICLE ACTUATED
	OR FIXED TIME

PERMANENTLY DISABLED MODES

MODES TO BE DISABLED	STREAM 0	STREAM 1	STREAM 2	STREAM 3
PART TIME (3)				
EMERGENCY VEHICLE				
HURRY CALL(S)				
SELECTED MANUAL CONTROL (3)				
URBAN TRAFFIC CONTROL				
SELECTED FIXED TIME OR VA				
CABLELESS LINKING				
BUS PRIORITY				
VEHICLE ACTUATED				
FIXED TIME				

✓ TICK IF REQUIRED TO BE DISABLED.

MANUAL FACILITY ENABLED	
ALWAYS	✓
WHEN HANDSET PLUGGED IN	
WHEN "MND" COMMAND ENTERED	

FACILITY SWITCH POSITION	NOT OPERATIVE INSERT (X)
VA	
FIXED TIME	
CLF	

(3) THESE MODES MAY ONLY BE DELETED ON ALL MODES SIMULTANEOUSLY

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

FIXED TIME MODE

Type of Fixed Time operation:	Option A - Fixed cycle running to current maximum

Option A - Fixed cycle running to current maximum - demand dependant phases

Phases	A	B	C	D																			
Dem' dep			✓																				

Option B - Fixed cycle running to timed stage sequence

Stage No.																							
Duration (sec)																							

Option C - Linked fixed time

Duration (sec)																								
Stream No.		Stage No.																						

Notes: Options B & C - Start up stages in each stream MUST be specified in the sequence.

Comments:

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

MANUAL PANEL

MANUAL SELECTION		
Button No.	Stage(s) called	Street Names - If Required
ALL RED	All Red Stage(s)	
1	1	Traffic
2	2	Peds
3	4	
4	5	
5	7	
6	8	
7		

ADDITIONAL SWITCHES		
Button No.	Description of Action	Momentary

* Momentary actions will only remain active whilst the associated button is pressed.

VISUAL INDICATORS / LED's	
Output No.	Function

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

URBAN TRAFFIC CONTROL

USE OF BITS								OTU MANUFACTURER & TYPE:-									
Bit No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Control	SO	F1	#F2	D2	F3	GO1							DS1	DS2	DS3	DS4	
Reply	MC	G1	G2	DR2	G3	DF	LF1	LF2									
Secret Sign																	Related Phase/Stage
Secret Sign with security																	Related Phase/Stage
Bit No	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
Control	SF1	SF2	SF3	SF4													
Reply					OC1	OC2	OC3	OC4	FL1	FL2	FL3	FL4					
Secret Sign																	Related Phase/Stage
Secret Sign with security																	Related Phase/Stage
Stream 1																	

0141 CONTROL AND REPLY BIT NOTATIONS

CONTROL BITS
Force (F1,F2,etc)
Stage Demand (D1,D2,etc)
Common Demand (DX)
SW FAC (SF1,SF2)
CLF Group Sync (SG)
RTC Sync (TS or TS1)
Fall Back Mode Select (FM)
Lamps On (LO)
Local Link Inhibit (LL)
Transmission Confirm (TC)
Solar Switch Over-ride (SO)
Pedestrian Demand (PX)
Hold Vehicle Stage (PV)
Solar Switch Over-ride (SO)

REPLY BITS
Stage Confirm (G1,G2,etc)
Stage Dem Confirm (DR1,DR2,etc)
SW FAC Confirm (SC1,SC2,etc)
CLF Group Sync Confirm (CS)
CLF Group 0 Confirm (GR1)
RTC Sync Confirm (CC or CC1)
Fall Back Confirm (FC)
Lamps Off Confirm (LE)
Hurry Call Confirm (HC)
Manual Control Mode (MC)
Controller Fault (CF)
Detector Fault (DF)
Remote Reconnect (RR)
One or more Signal Lamps Failed (LF)
Pedestrian Inhibit by Lamp Monitor (PI)
Pedestrian Reply (GX)

UTC CONTROL REQUIRED WHEN:-

✓ Force Bits Present
Force Bits and TC/TO Present
TC/TO Present

B.T. Termination requirements are:-

B.T. Line Number:-

ORN:-	397L	Scheme Number:-	0	Issue:-	1	Date:-	04/12/2024

OPTION 1 (MCE 0105/0106)
OPTION 2 (TCD316) (delete one)

INSERT PHASES OR STAGES TO BE DEMANDED, IF DEMANDS TO BE UNLATCHED ADD #										
D2										
C										
INSERT PHASES TO BE EXTENDED										
INSERT PHASES OR STAGES FOR REPLY										
DR2										
2										
DEMAND DEPENDENT FORCES INSERT DEMANDS TO BE CONSIDERED										
F2										
C										

NOTES:-
1. Hash (#) indicates demand dependent force
2. Asterisk (*) denotes Control or Reply associated with separate Pelican

UTC ONLY - MANUAL MODE, NO LAMP SUPPLY, LAMP FAILURES etc., CONFIRM					
TICK AS APPROPRIATE:-	G1/G2	MC	DF	LF1	LF2
MANUAL MODE SELECTED AND OPERATIVE	✓	✓			
NO LAMP POWER and/or SIGNALS OFF	✓				
DETECTOR FAULT			✓		
SELECT SWITCH NOT IN NORMAL POSITION		✓			
ANY LAMP FAILURE				✓	
2nd RED LAMP FAILURE ON ANY PHASE				✓	✓

Leeds UTC Lamp Monitoring Specification:

On any lamp failure the LF1 reply bit shall be returned.

On the 1st red lamp failure to any phase conflicting with a pedestrian phase, the LF1 reply bit shall be returned and the detector fault monitor lamp shall flash.

On any second red lamp failure to any phase LF2 shall be returned.

On any second red lamp failure to any phase conflicting with a pedestrian phase, the LF2 reply bit shall be sent, the detector fault monitor lamp shall continue to flash, the conflicting pedestrian phase wait indicators shall be illuminated and the conflicting pedestrian phase demands shall be inhibited (except in the case of a pedestrian crossing, where the signals will be extinguished). All other non-conflicting pedestrian phases shall continue to be serviced.

On any lamp failure there shall be no intergreen extensions.

Master Time Clock

ID.	Day Type	Hours	Mins	Secs	Introduce Function Required	Function Number	Plan / Parameter
1	9	07	00	00	Max Set A	2b	
2	9	09	15	00	Max Set B	2b	
3	9	15	00	00	Max Set C	2b	
4	9	18	30	00	Max Set D	2b	
5	9	21	00	00	Max Set E	2b	
6	8	00	00	00	Max Set F	2b	
7	6	09	30	00	Max Set A	2b	
8	6	15	00	00	Max Set G	2b	
9	6	18	30	00	Max Set H	2b	
10	6	21	00	00	Max Set E	2b	
11	7	09	30	00	Max Set A	2b	
12	7	15	00	00	Max Set G	2b	
13	7	18	30	00	Max Set H	2b	
14	7	21	00	00	Max Set E	2b	
15	8	07	00	00	Enable audibles		
16	8	22	00	00	Disable audibles		
17							
18							
19							
20							
21							
22							
23							
24							
25							
26							
27							
28							

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

Type of day

1 Monday
2 Tuesday
3 Wednesday
4 Thursday
5 Friday
6 Saturday
7 Sunday
8 All Week
9 All Week, except Sat & Sun
10 All Week, except Sun

For complex day selections use day numbers, e.g. Mon & Wed = 1, 3

Function Numbers

0 Isolate controller
1 Introduce plan
2 Introduce event defined below:
- a Switch On Input/Output Active / Inactive / Normal
- b Introduce Standard / Alternative Max Setting
- c Switch a sign On / Off
- d Switch a Phase / Stage In / Out of cycle
- e Switch To / From Part Time mode

Cableless Linking Facility To be made available

Name				Name				Name				Name			
Plan No.	Stream No.	Offset	Cycle Time	Plan No.	Stream No.	Offset	Cycle Time	Plan No.	Stream No.	Offset	Cycle Time	Plan No.	Stream No.	Offset	Cycle Time
Group Number	Group Time	Group Influence	Related Stage	Group Number	Group Time	Group Influence	Related Stage	Group Number	Group Time	Group Influence	Related Stage	Group Number	Group Time	Group Influence	Related Stage

Notes: Group Influence **A** - Isolate to VA (Stage number not required), **B** - An Immediate move to stage "N", **C** - A demand dependant move to stage "N", **D** - Hold (No stage change to occur), **E** - Prevent moves except to stage "N".

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLELESS LINKING	
STREAM 1	
1 - 2	(($\overline{F}$1 • F2) + (F1 • F2 • GO • $\overline{EXT}$ A • $\overline{EXT}$ B)) • (D2 + DEM C)
1 - 3	($\overline{F}$1•F3)
2 - 1	($\overline{F}$2•F1)
2 - 3	($\overline{F}$2•F3)
3 - 1	($\overline{F}$3•F1)
3 - 2	($\overline{F}$3•F2) • (D2 + DEM C)

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

$\overline{EXT}$ C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED "." - BOOLEAN "AND" "+" - BOOLEAN "OR"

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024

SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLELESS LINKING	
MOVA STREAM 1	
1 - 2	$\overline{F}$1•F2•MTO1
2 - 1	$\overline{F}$2•F1•MTO1
1 - 3	$\overline{F}$1•F3•MTO1
3 - 1	$\overline{F}$3•F1•MTO1
2 - 3	$\overline{F}$2•F3•MTO1
3 - 2	$\overline{F}$3•F2•MTO1

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

$\overline{EXT}$ C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED "." - BOOLEAN "AND" "+" - BOOLEAN "OR"

ORN:- 0	Scheme Number:- 0	Issue:- 1	Date:- 00/01/1900

SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- VEHICLE ACTUATED / FIXED TIME	
STREAM 1	
1 - 2	(MAX A + $\overline{EXT}$ A) • (MAX B + $\overline{EXT}$ B) • (DEM C)
1 - 3	
2 - 1	$\overline{MIN}$ C
2 - 3	
3 - 1	(DEM A + DEM B)
3 - 2	(DEM C)

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

$\overline{EXT}$ C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED "." - BOOLEAN "AND" "+" - BOOLEAN "OR"

ORN:- 397L	Scheme Number:- 0	Issue:- 1	Date:- 04/12/2024